

Donghyun Son

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EDUCATION

Stanford University, M.S. in Computer Science

Sep 2026 — Present

Seoul National University, B.S. in Computer Science

Mar 2018 — Feb 2026

GPA: 4.04/4.3 (4.11/4.3 in major), *Summa Cum Laude*

University of Texas at Austin, Exchange Student, Computer Science

Sep 2025 — Dec 2025

PUBLICATIONS

Research Keywords: ML Systems, Quantization, KV Cache Compression, Domain Generalization

*: equal contribution.

[TR1] *K-EXAONE 2.0 Technical Report*

LG AI Research

[W2] *EntropyCache: Decoded Token Entropy Guided KV Caching for Diffusion Language Models*

Minsoo Cheong, Donghyun Son, Woosang Lim, Sungjoo Yoo

AdaptFM: Resource-Adaptive Foundation Model Inference @ ICML 2026 (AdaptFM @ ICML 2026)

[P1] *NALAR: A Serving Framework for Agent Workflows*

Benedict Marco Laju, Donghyun Son, Saurabh Agarwal, Nitin Kedia, Myungjin Lee, Jayanth Srinivasa, Aditya Akella

Preprint, Under Review

[C2] *NSNQuant: A Double Normalization Approach for Calibration-Free Low-Bit Vector Quantization of KV Cache*

Donghyun Son, Euntae Choi, Sungjoo Yoo

Advances in Neural Information Processing Systems (NeurIPS 2025)

[W1] *Gradient Estimation for Unseen Domain Risk Minimization with Pre-Trained Models*

Byounggyu Lew*, Donghyun Son*, Buru Chang

Workshop and Challenges for Out-of-Distribution Generalization in Computer Vision @ ICCV 2023 (OOD-CV@ICCV 2023)

[C1] *Reliable Decision from Multiple Subtasks through Threshold Optimization: Content Moderation in the Wild*

Donghyun Son*, Byounggyu Lew*, Kwanghee Choi*, Yongsu Baek, Seungwoo Choi, Beomjun Shin, Sungjoo Ha, Buru Chang

ACM International Conference on Web Search and Data Mining (WSDM 2023, Oral)

WORK EXPERIENCE

AI Scientist Intern, EXAONE Lab

Apr 2026 — Jul 2026

LG AI Research

Seoul, South Korea

- Integrated and optimized EXAONE models across LLM serving frameworks (vLLM, SGLang)
- Implemented EXAONE model support for Ollama, including MTP-based speculative decoding
- Investigated reinforcement learning pipelines to identify sources of training-inference mismatch

Undergraduate Researcher

Aug 2025 — Dec 2025

UTNS Research Group @ UT Austin, Supervised by Prof. Aditya Akella

Austin, Texas

- Designed and implemented an agentic system where the central controller dynamically manages request migration, load balancing, and scaling based on a global observability (under review)

Undergraduate Research Intern

Sep 2024 — Aug 2025

CMALab @ SNU, Supervised by Prof. Sungjoo Yoo

Seoul, South Korea

- Investigated a calibration-free vector quantization method for KV cache, incorporating a novel normalization algorithm and efficient CUDA kernels for 1–2 bit inference (NeurIPS25)

Machine Learning Engineer, Moderation ML Team

Mar 2022 — Jul 2023

Hyperconnect (Acquired by Match Group)

Seoul, South Korea

- Built an ML-based system for moderating images and audio across Match Group brands (e.g., Tinder, Hinge)
- Devised a threshold optimization algorithm based on GD to optimize the tradeoff between reliability and cost (WSDM23, Oral)
- Proposed a training algorithm for ML models to generalize effectively across different services and domains (OOD-CV@ICCV23)

- Designed and implemented C++-based multi-threaded ML inference pipeline for low-latency gaze estimation
- Built a robust training procedure to train an appearance-based gaze estimation model and improved accuracy by 30%
- Proposed a transformer-based calibration algorithm that predicts user-specific latent vectors using users' samples

ACADEMIC SERVICES

Conference Reviewer: NeurIPS 2026

Workshop Reviewer: AdaptFM@ICML 2026

PATENTS

Apparatus and Method for Setting Criteria on Data Classification (US20240281494A1)

Yong Su Baek, **Dong Hyun SON**, Beom Jun Shin, Byoung Gyu Lew, Bu Ru CHANG, Kwang Hee CHOI, Seung Woo Choi, Sung Joo Ha

Apparatus for Domain Generalization of Machine Learning Models, Methods and Computer Readable Recording Mediums Therefor (US20240087294A1)

Bu Ru CHANG, Byoung Gyu Lew, **Dong Hyun SON**

HONORS & AWARDS

Programming Contests

- **2nd prize**, at Union of Clubs Programming Contest (UCPC) 2025
- **5th prize**, at Samsung Collegiate Programming Contest (SCPC) 2021
- **5th place**, at ICPC NERC Huawei Challenge 2020: Cloud Scheduling Challenge
- **1st place**, at Seoul National University Programming Contest (SNUPC) 2019, div. 2

Scholarships

- **Study Abroad Scholarship of Kwanjeong Educational Foundation** (\$60,000), Kwanjeong Educational Foundation
- **Samsung Software Membership**, from Samsung Research (Nov 2021 — Feb 2026)
- **Korea-U.S. STEM Student Exchange Scholarship** (\$9,000), from Ministry of Trade, Industry and Energy (2025.09)
- **Merit-based Full Tuition Academic Scholarship** (\$2,300), from Seoul National University (2020-1, 2023-2, 2024-1, 2024-2)

Others

- **4th place**, at CXR-LT Challenge Task 1 (MICCAI 2024)
- **Bronze Prize**, at 42nd Undergraduate Mathematics Contest div.1 (for mathematical majors)
- **Microsoft Azure Champ Prize : Hack For Good**, at TartanHacks 2021 in CMU

PRESENTATIONS

- **KV Cache Compression for Long Context Inference**, Weekly Seminar @ Deepest S16 (January 2025)
- **Efficient Algorithms for LLM Inference**, Weekly Seminar @ Deepest S15 (July 2024)

ADDITIONAL INFORMATION

References: Prof. Sungjoo Yoo (Seoul National University), Prof. Buru Chang (Korea University), Dr. Saurabh Agarwal (UT Austin)

Interests: Table Tennis, Math Puzzles, Mind Sports (4 Dan in Go)

Technical Skills: C++, Python, Pytorch, Tensorflow, Jax, CUDA, OpenCL, MPI, OpenMP

Problem Solving: Codeforces (handle: **diordhd**, rating: 2200+) / BOJ (handle: **dhdroid**)

TOEFL iBT: 108 (R29/L29/S22/W28)